

LECTURE-18

Thursday
16/4/26

☐ Deadlock Prevention: Violate any of four condition of deadlock.

1. Make all ~~resources~~ resources shareable (like CPU for some time).

But Printer cannot be shareable at all.

2. No Hold & wait: Ask a process before execution which resources does it wants and give all of them to it at once at start so it will hold all at once & no need to wait (Resource waste).

OR whenever a new resource req. by any process; it will be bound to release any previous resource held (problem if process needs both resources at once).

3. Allow Pre-emption: If process has a resource A and request for a Busy resource B ^{and waits} then we'll pre-empt and take resource A as well from that process bcs wait to usnay karna hai to release resource A.

4. Circular wait: Assign integer no. to resources in increasing order. Then if a process has 'resource 3' and wants 'resources 4'; then it will get it and no problem. But if 'Resource 3' wants 'resource 2', then it can't have it until it releases resource 3. This way no cycle will begin and only series will be there. (List: Higher No. resources isn't allowed to req. lower no. resource).

☐ Deadlock Avoidance:

We'll take info from process before execution which resources & how many instances of it does process needs. This way CPU can pre-plan allocation of resources so no deadlock.

→ Safe State:

State which have at least one safe sequence.

→ Safe Sequence:

Available resources hotay waqt; agar ham ek specific sequence nikal lein jis say saray processes ko required resources mil jaye; that sequence is safe sequence.

Q: System with 12 tape drives & 3 process

Process	Max Need	Allocated
P0	10	5
P1	4	2
P2	9	2

9 tapes in used processes
 $12 - 9 = 3$ remaining

* Now in 3 remaining tapes which are free; we need to find a sequence jismay ham aagay pechay kr kay saray processes ki demand puri ho.

P1 → P0 → P2 ← Safe Sequence.

- 2/3 tapes given to P1 when it only required $4 - 2 = 2$ tapes
- P1's max need is fulfilled as all tapes given i.e.: 4.
- Now we've 4 resources/tapes released from P1 and ek pehlay ka bach gaya tha in ①. ∴ Current tapes = $4 + 1 = 5$.
- 5 tapes needed by P0 and we also have 5; so given to P0.
- Now 10 resource/tapes released by P0
- P2 requires 7 tapes so given & ALL sorted.

* If at ①; P1 wld've asked for 3 tapes which were available in CPU; we still won't give as it causes a deadlock afterwards at some point.

→ Resource Allocation Graph: (Continue)

• Claim Edge:→

(P1)→ [R1]

- Since we've now told CPU which resources is needed by any process (in the beginning), therefore we can predict with dotted line that it process may ask for this resource.

- When that hypothetical claim req. is actually requested; then we'll see if allowing that resource will not make a cycle; only then we'll give that claim resource back to process.